

# Econometrics I

## Review: Hypothesis Testing for Population Means

Standalone version from the original review

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## Estimation of the Population Mean

This set of slides are the stand alone version of the section about hypothesis testing in the original Review of Probability and Statistics slides. I put this version for your convenience.

- Now you have enough knowledge to describe the process of estimating a population mean with more formality.
- Consider population described by the random variables  $Y$  and  $\epsilon$  and suppose that we have access to a random sample of size  $n$  of such population. The observations satisfy:

$$y_i = \mu_Y + \epsilon_i, i = 1, \dots, n$$

where the collection of random variables  $(y_i, \epsilon_i)$   $i = 1, \dots, n$  are independent and identically distributed, since they come from a random sample from the same parent population.

- Furthermore, suppose that the following assumptions are true for  $\epsilon$  :

1  $E(\epsilon) = 0$

2  $Var(\epsilon) = \sigma^2$

- Then  $E(Y) = \mu_Y$  and  $\sigma_Y^2 = Var(Y) = \sigma^2$ .
- The previous model is actually a (simple) linear regression model that satisfies the Gauss-Markov assumptions.

The Gauss-Markov theorems allows us to conclude that, for the previous population model:

### Properties of $\bar{Y}$

- The sample mean  $\bar{Y}$  is an unbiased estimator of  $\mu_Y$ .
- The sample mean  $\bar{Y}$  is the best linear unbiased estimator (BLUE), where “best” stands for more efficient here.
- The sample mean  $\bar{Y}$  is also the least squares estimator of  $\mu_Y$ .

# Hypothesis Tests Concerning the Population Mean

## Hypothesis Tests About $\mu_Y$

The objective is to make a provisional decision based on the evidence at hand whether a *null* hypothesis is true, or instead that some *alternative* hypothesis is true. That is, test

1  $H_0 : E[Y] = \mu_{Y,0}$  versus  $H_I : E[Y] > \mu_{Y,0}$  (1-sided  $>$ ).

2  $H_0 : E[Y] = \mu_{Y,0}$  versus  $H_I : E[Y] < \mu_{Y,0}$  (1-sided  $<$ ).

3  $H_0 : E[Y] = \mu_{Y,0}$  versus  $H_I : E[Y] \neq \mu_{Y,0}$  (2-sided).

Here,  $\mu_{Y,0}$  denotes the hypothesised value of the population mean (i.e., the value of the population mean “under” the null hypothesis).

# Hypothesis Tests Concerning the Population Mean

## The *p-value*

The p-value is the probability of drawing a statistic at least as adverse to the null hypothesis as the value actually computed with your data, assuming that the null hypothesis is true. One often “rejects the null hypothesis” when the p-value is less than the significance level  $\alpha$ .

## The *significance level* $\alpha$

The significance level of a test is a pre-specified probability of incorrectly rejecting the null, when the null is true.


$$\text{Pr. type I error : } \Pr(\text{Rejecting the Null} \mid H_0 \text{ true}) = \alpha$$

## Hypothesis Tests Concerning the Population Mean

### Example

$$H_0: \mu = 1.75 \quad \text{vs} \quad H_1: \mu \neq 1.75$$

- Suppose we are analyzing the height of the population of students at Nazarbayev University, and that we believe the average height is 1.75 mts. This is our null hypothesis.
- We draw a random sample and we find that the average height in the sample is 1.8 mts.
- The p-value is the probability of observing a sample mean  $\bar{Y} = 1.8$  mts. when the true population average is 1.75 mts.
- If the p-value is small (for example ~~0.05~~), then it is very unlikely that a random sample would lead to an estimate of 1.8 mts when the true population value is 1.75 mts.

## Hypothesis Tests Concerning the Population Mean

### Calculation of the *p-value*

Let  $\bar{Y}^{act}$  be the value of the sample average “actually” computed with the sample data and let  $Pr_{H_0}$  denote the probability computed under the null hypothesis (i.e., computed assuming that  $E[Y] = \mu_{Y,0}$ ). Then, the p-value is

$$\text{p-value} = Pr_{H_0} \left( |\bar{Y} - \mu_{Y,0}| > |\bar{Y}^{act} - \mu_{Y,0}| \right) \quad \text{two-sided} \quad (1)$$

$$\text{p-value} = Pr_{H_0} \left( (\bar{Y} - \mu_{Y,0}) > (\bar{Y}^{act} - \mu_{Y,0}) \right) \quad \text{one-sided} \quad (2)$$

## Hypothesis Tests Concerning The Population Mean

### Computing the p-value for $n$ Large and $\sigma_Y$ Known

$$\begin{aligned}\text{p-value} &= Pr_{H_0} \left( \left| \bar{Y} - \mu_{Y,0} \right| > \left| \bar{Y}^{act} - \mu_{Y,0} \right| \right), \\ &= Pr_{H_0} \left( \left| \frac{\bar{Y} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= Pr_{H_0} \left( Z_0 > \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right), \\ &= 2\phi \left( - \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{\sigma_{\bar{Y}}} \right| \right).\end{aligned}\tag{3}$$

where  $\sigma_{\bar{Y}} = \sigma_Y / \sqrt{n}$ .

## Hypothesis Tests Concerning The Population Mean

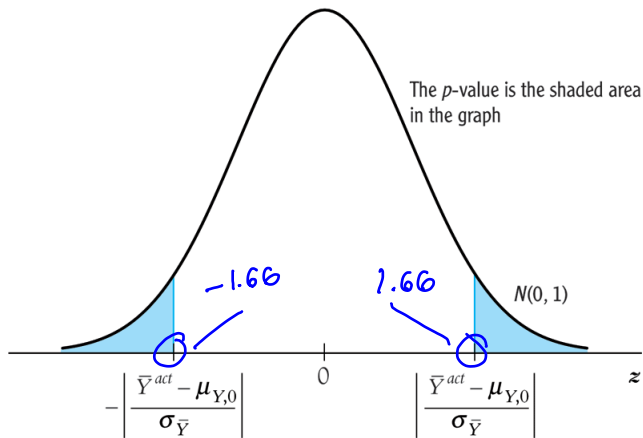


Figure:  $p$ -value for known variance.

## Hypothesis Tests Concerning the Population Mean

### Example

Under the null  $H_0$

$$T = \frac{\bar{Y} - 1.75}{\frac{0.3}{\sqrt{100}}} \sim N(0, 1)$$

- Suppose we know that the standard deviation in the distribution of height among NU students is 0.3 mts (i.e.,  $\sigma_Y = 0.3$ ).
- In a sample of 100 students the computed sample average height is  $\bar{Y} = 1.8$  mts.
- Compute the p-value for the tests
  - 1  $H_0 : E[Y] = \mu_{Y,0} = 1.75$  versus  $H_I : E[Y] > 1.75$  (1-sided  $>$ ).
  - 2  $H_0 : E[Y] = \mu_{Y,0} = 1.75$  versus  $H_I : E[Y] \neq 1.75$  (2-sided).

$$T^0 = \frac{1.8 - 1.75}{\frac{0.3}{\sqrt{100}}} = \frac{0.05}{\frac{0.3}{10}} = \frac{0.05}{0.03} = 1.666$$

## Hypothesis Tests Concerning The Population Mean

### Remarks

- However, to proceed in the way we suggested above, we still need to know the population variance.
- In general the population variance is unknown and it must be estimated before the p-value can be computed.

## Hypothesis Tests Concerning the Population Mean

### Sample Variance and Sample Standard Deviation

The sample variance  $s_y^2$  is

$$s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2. \quad (4)$$

The sample standard deviation  $s_Y$  is the square root of the sample variance. The sample variance is an unbiased and consistent estimator of the population variance.

## Hypothesis Tests Concerning the Population Mean

### Standard Error of $\bar{Y}$

The standard error of  $\bar{Y}$  is an estimator of the standard deviation of  $\bar{Y}$  and is denoted by  $SE(\bar{Y})$  or  $\bar{s}$ . When  $Y_1, \dots, Y_n$  are i.i.d. we have

$$SE(\bar{Y}) = \bar{s} = \frac{s_Y}{\sqrt{n}}. \quad (5)$$

## Hypothesis Tests Concerning The Population Mean

### Computing the p-value for $n$ Large and $\sigma_Y$ Unknown

Since the sample variance is a consistent estimator of the population variance, the p-value can be computed by using  $SE(\bar{Y})$ . Thus, when  $\sigma_Y$  is unknown we can use the formula

$$\text{p-value} = 2\phi \left( - \left| \frac{\bar{Y}^{act} - \mu_{Y,0}}{SE(\bar{Y})} \right| \right), \quad (6)$$

where  $SE(\bar{Y}) = \bar{s} = \frac{s_Y}{\sqrt{n}}$ .

## Hypothesis Tests Concerning The Population Mean

### The t-statistic

The standardized sample average  $(\bar{Y} - \mu_{Y,0}) / SE(\bar{Y})$  plays an important role in testing statistical hypotheses and it is called the *t-statistic* or *t-ratio*,

$$t = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}. \quad (7)$$

When  $n$  is large, the sample variance is close to the population variance, and thus the distribution of the t-statistic can be approximated by the standard normal.

## Hypothesis Tests Concerning The Population Mean

### The t-statistic and the p-value

We can write the formula for the p-value in terms of the t-statistic. Let  $t^{act}$  denote the value of the t-statistic actually computed

$$t^{act} = \frac{\bar{Y}^{act} - \mu_{Y,0}}{SE(\bar{Y})}. \quad (8)$$

Hence, when  $n$  is large the p-value can be computed by

$$\text{p-value} = 2\phi(-|t^{act}|) \quad \text{two-sided}, \quad (9)$$

$$\text{p-value} = 1 - \phi(t^{act}) \quad \text{one-sided } >, \quad (10)$$

$$\text{p-value} = \phi(t^{act}) \quad \text{one-sided } <. \quad (11)$$

# Hypothesis Tests Concerning the Population Mean

## Example

- Suppose the standard deviation in the distribution of height among NU students is unknown ( $\sigma_Y$  unknown).
- In a random sample of 100 students the computed sample average height is  $\bar{Y}^{act} = 1.8$  mts and the sample variance is  $s^2 = 0.08$ .
- Compute the p-value for the test  $H_0 : E[Y] = \mu_{Y,0} = 1.75$  versus  $H_I : E[Y] > 1.75$ .

# Hypothesis Tests Concerning the Population Mean

## Hypothesis Testing With a Prespecified Significance Level

### Type I Error

Reject the null hypothesis when it is true.

### Type II Error

Not rejecting the null hypothesis when it is false.

### Significance Level

The significance level of the test is the probability of rejecting the null hypothesis when it is true (probability of type I error). Hence, another way to think about the p-value is as the smallest significance level at which you can reject the null hypothesis.

# Hypothesis Tests Concerning the Population Mean

## Hypothesis Testing With a Prespecified Significance Level

### Critical Value

The critical value of the test is the value of the test statistic for which the test just rejects the null hypothesis at the chosen significance level.

### Rejection Region

The rejection region is the set of values of the test statistic for which the null hypothesis is rejected.

### Acceptance Region

The acceptance region is the set of values of the test statistic for which the null hypothesis is not rejected.

# Hypothesis Tests Concerning the Population Mean

## Hypothesis Testing With a Prespecified Significance Level

### Size of the Test

The size of the test is the probability of type I error.

### Power of the Test

The power of the test is the probability of rejecting the null hypothesis when the alternative is true.

# Hypothesis Tests Concerning the Population Mean

## Hypothesis Testing With a Prespecified Significance Level

### How Do We Choose the Significance Level?

- Econometricians typically use a 5% significance level, which means that they are incorrectly rejecting the null hypothesis on average 1 out of 20 times.
- Using a very low significance level makes more difficult to reject the null hypothesis when it is false (i.e., the lower the significance level the lower the power of the test).

# Hypothesis Tests Concerning the Population Mean

## Implementing a Two-Sided Test on the Population Mean

- 1 Compute  $SE(\bar{Y})$  (standard error of  $\bar{Y}$ ).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0}(Z > |t^{act}|) = 2\phi(-|t^{act}|).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if  $|t^{act}| > 1.96$ , where 1.96 is the critical value of the  $t$  statistic (i.e., the value that leaves 2.5% of the probability mass to the right).

# Hypothesis Tests Concerning the Population Mean

## Implementing a One-Sided ( $>$ ) Test on the Population Mean

- 1 Compute  $SE(\bar{Y})$  (standard error of  $\bar{Y}$ ).
- 2 Compute the t-statistic.
- 3 Compute the p-value as

$$\text{p-value} = Pr_{H_0} (Z > t^{act}) = 1 - \phi(t^{act}).$$

- 4 Reject the null hypothesis at the 5% significance level if the p-value is less than 0.05.
- 5 Equivalently, reject the null hypothesis if  $t^{act} > 1.64$ , where 1.64 is the critical value of the  $t$  statistic (i.e., the value that leaves 5% of the probability mass to the right).

## Confidence Intervals for the Population Mean

- Even though we cannot learn the true value of  $\mu_Y$  (the population mean of  $Y$ ) using only a sample, we can construct an interval containing  $\mu_Y$  with a certain prespecified probability.
- This interval is called *confidence interval*, and the probability level is called the *confidence level*.
- For example, a 95% confidence interval for  $\mu_Y$  is an interval that contains the true value of  $\mu_Y$  in 95% of repeated samples.

## Confidence Intervals for the Population Mean

### Some Confidence Intervals

When sample sizes are large the 90%, 95%, and the 99% confidence intervals are

$$90\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.64 \times SE(\bar{Y})] . \quad (12)$$

$$95\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 1.96 \times SE(\bar{Y})] . \quad (13)$$

$$99\% \text{ confidence interval for } \mu_Y : [\bar{Y} \pm 2.58 \times SE(\bar{Y})] . \quad (14)$$

## Hypothesis Tests Concerning the Population Mean

### Remarks: What is Random Here?

- The values of  $Y_1, \dots, Y_n$  and thus any function of them – including the confidence interval which will differ from one sample to the next.
- The population parameter  $\mu_Y$  is not random; we just don't know it.

## Using the t-Statistic When the Sample Size is Small

### The t-Statistic Testing the Mean

Let  $Y_i$  for  $i = 1, \dots, n$  be an i.i.d. sample. As we have stated before, the t-statistic for testing  $H_0 : E(Y) = \mu_{Y,0}$  is

$$t = \frac{\bar{Y} - \mu_{Y,0}}{\sqrt{s_Y^2/n}} = \frac{\bar{Y} - \mu_{Y,0}}{SE(\bar{Y})}, \quad (15)$$

where  $s_Y^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$  is the sample variance.

## Using the t-Statistic When the Sample Size is Small

### The t-Statistic Testing the Mean

- When the sample size is small, the standard normal distribution is a poor approximation for the distribution of the t-statistic.
- The exact distribution of the t-statistic is complicated and depends on the distribution of  $Y$ .
- However, if the population distribution is normal, the exact distribution of the t-statistic testing the mean of a single population is the *Student t* distribution with  $n - 1$  degrees of freedom.

## Using the t-Statistic When the Sample Size is Small

### Some Remarks on the Use of the Student $t$ Distribution in Practice

- For testing the mean of  $Y$ , the *Student  $t$  Distribution* can be used **only** if the underlying population distribution is normal.
- If the sample size is moderate (several dozen) or large (hundreds or more) the difference between the  $t$ -Distribution and the Standard Normal critical values is negligible.

## Hypothesis Tests Concerning The Population Mean

| degrees of freedom<br>( $n - 1$ ) | 5% $t$ -distribution<br>critical value |
|-----------------------------------|----------------------------------------|
| 10                                | 2.23                                   |
| 20                                | 2.09                                   |
| 30                                | 2.04                                   |
| 60                                | 2.00                                   |
| $\infty$                          | 1.96                                   |

Figure: Critical values in the  $t$ -Student Distribution

## Using the t-Statistic When the Sample Size is Small

### Some Remarks on the Use of the Student t Distribution in Practice

- In sum, the Student-t distribution is only relevant when the sample size is very small; but in that case, for it to be correct, you must be sure that the population distribution of  $Y$  is normal.
- In economic data, the normality assumption is rarely credible.

# Using the t-Statistic When the Sample Size is Small

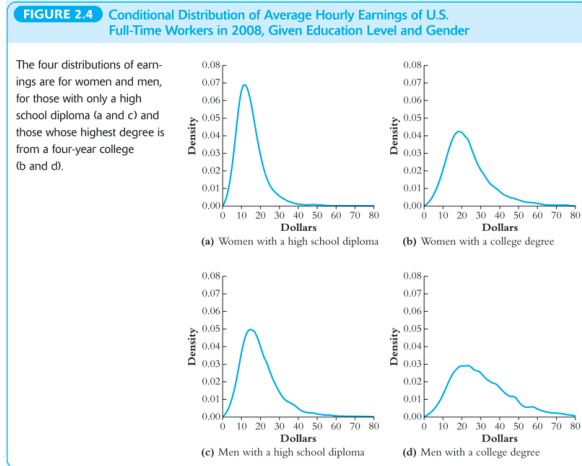


Figure: Economic Data and the Normality Assumption